Cross-Variety Initialization and Alignment-Supervised Durations for One-Hour Hakka Speech Synthesis

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***Abstract*—Hakka is a tonal language with several regional varieties, many of which have limited recorded speech resources. This paper investigates neural text-to-speech (TTS) for two Taiwanese Hakka varieties, Sixian and Hailu, under a one-hour target-variety training condition. With limited training data, learned alignments may result in inaccurate segment durations and unreliable tone realization. This paper presents XAD-TTS, a Matcha-TTS training recipe that combines cross-variety initialization with forced-alignment duration supervision and an enlarged duration predictor. The target model is initialized from a model trained on a related Hakka variety. Across three training seeds, XAD-TTS increases the proportion of checked-syllable vowel nuclei whose durations, measured on synthesized waveforms, are within 30 ms of the corresponding natural durations from 71.0% to 81.3% for Sixian and from 69.4% to 82.2% for Hailu. For this duration measure, the results are close to those of full-data baselines trained on 48.0 and 47.2 hours of target-variety speech, respectively. XAD-TTS also reduces mel-cepstral distortion by 0.33 dB for Sixian and 0.24 dB for Hailu, while improving objective tone-classification accuracy. An additional automatic speech recognition analysis shows improved agreement for checked-syllable codas and tone marks relative to the transcripts of natural recordings.**

***Index Terms*—duration modeling, forced alignment, Hakka, low-resource languages, speech synthesis, transfer learning**

I. Introduction

Hakka is a Sinitic tonal language spoken in several regional varieties. In Taiwan, Sixian and Hailu are two major Hakka varieties. They share many consonants and vowels but differ in their tone systems. Although Hakka speech technology has begun to emerge [1], recorded speech resources remain limited for individual varieties, making it difficult to develop TTS systems when only a small amount of target-variety data is available.

Neural TTS models such as Matcha-TTS [2] learn alignments between input symbols and acoustic frames during training, typically using monotonic alignment search (MAS) [3]. A duration predictor is then trained using the resulting alignments. Under limited-data conditions, alignment and duration prediction may be less reliable. In Hakka, this issue is particularly relevant to checked syllables, which end in an unreleased stop, written as *-b*, *-d*, or *-g* in the romanization used in this study. These syllables typically have relatively short vowel nuclei; therefore, duration is one of the acoustic cues relevant to their realization.

Previous studies have explored approaches to low-resource TTS, including initialization from models trained on related, better-resourced languages [4], [5], and the use of durations obtained from an external aligner, as in FastSpeech 2 [6] with the Montreal Forced Aligner (MFA) [7]. This paper combines cross-variety initialization and alignment-supervised duration modeling for Hakka TTS, focusing on conditions with limited target-variety training data. The main contributions are as follows:

- **XAD-TTS:** XAD-TTS, short for cross-variety initialization with alignment-supervised durations, is a Matcha-TTS training recipe that initializes a target-variety model from a related source variety and supervises an enlarged duration predictor using forced-alignment durations. The aligner is not required during synthesis, although it is used to obtain training targets and to measure durations during evaluation.
- **Experiments across data sizes:** Evaluation on Sixian and Hailu using three training seeds at 1 h and 10 h, together with single-seed full-data baseline references at 48.0 h and 47.2 h.
- **Evaluation beyond spectral distance:** Analysis of synthesized vowel durations, objective tone classification, and ASR-based agreement between synthesized and natural speech.

II. Method

A. Baseline model

The baseline system, denoted Base, uses the original Matcha-TTS architecture [2] and is trained from scratch.

Matcha-TTS encodes input symbols into per-symbol features and predicts their durations. The features are expanded according to the durations, and a conditional flow-matching decoder [8] generates a mel spectrogram. A HiFi-GAN vocoder [9] then converts the mel spectrogram into a waveform.

During training, MAS estimates the alignment between input symbols and mel frames, and the duration predictor is trained using the resulting durations. The training objective consists of duration, prior, and flow-matching losses. In the original inference procedure, predicted durations are rounded up to whole frames. The rounding procedure considered in this study is described in Section II-D.

B. Cross-variety initialization

A source model is trained from scratch on the full training pool of the other Hakka variety: the 47.2 h Hailu model initializes a Sixian target, and the 48.0 h Sixian model initializes a Hailu target. Sixian and Hailu are closely related varieties, and both use a shared symbol table. For target-model initialization, parameters with matching names and shapes are copied from the source model, except for the speaker embeddings and, in XAD-TTS, the enlarged duration predictor. These components are re-initialized, no parameters are frozen, and the optimizer is restarted.

Training data are normalized with statistics calculated from the target-variety training data. Because the copied model buffers contain the mel-normalization statistics of the source variety, synthesis also uses the target-variety statistics.

C. Alignment-supervised durations

For each variety, an MFA acoustic model was trained on all HAT TTS recordings of that variety and their transcripts, including the validation and test recordings, before the data were split, using a pronunciation lexicon derived from the romanization by deterministic rules. MFA training uses only audio and transcripts, without duration labels, and the same alignments provide the natural reference durations used in Section III-C. Over 99% of the utterances were successfully aligned; the remaining 42 Sixian and 143 Hailu utterances were excluded from all subsets. The resulting phone intervals are converted into frame counts using a hop size of 256 samples at a sampling rate of 22.05 kHz, corresponding to approximately 11.6 ms per frame.

Matcha-TTS inserts blank tokens between input symbols. Silence between two phones is assigned to the pause symbol at that position if the text contains one, and otherwise to the blank token between the two phones. Every symbol, including each blank, receives at least one frame, taken from the preceding symbol when possible. The resulting durations replace MAS-derived durations during training. They serve as targets for the duration loss and determine the feature expansion used in the prior and decoder losses.

D. Larger duration predictor and rounding

To increase the capacity of the duration predictor, its kernel size is changed from 3 to 5 and its number of channels from 256 to 512. When cross-variety initialization is applied in XAD-TTS, the enlarged duration predictor and speaker embeddings are re-initialized.

Rounding every predicted duration up can increase the total predicted duration. To reduce the potential effect of rounding, each trained model uses either rounding up or cumulative rounding (rounding the running sum, which preserves the total length). The rule is chosen on the validation set only, and the test set is not used for this choice: the rule whose rounded predicted vowel-nucleus durations have the smaller mean absolute error with respect to the MFA-derived durations of the validation sentences, averaged over checked and other syllables, is selected. The same selection procedure is applied to all systems, including Base.

XAD-TTS combines cross-variety initialization (Section II-B), alignment-supervised durations (Section II-C), and the enlarged duration predictor (Section II-D); its name refers to cross-variety initialization (X) and alignment-supervised durations (AD). The rounding selection is not specific to XAD-TTS, since it is applied to every system. Fig. 1 illustrates the training, synthesis, and evaluation procedures. The target model is initialized from the source-variety model and trained on the one-hour target training speech and text, while MFA-derived durations from the same data provide supervision for the enlarged duration predictor and alignment information for training. After training, the model synthesizes speech from test text. The synthesized speech is then evaluated using natural test speech as a reference.

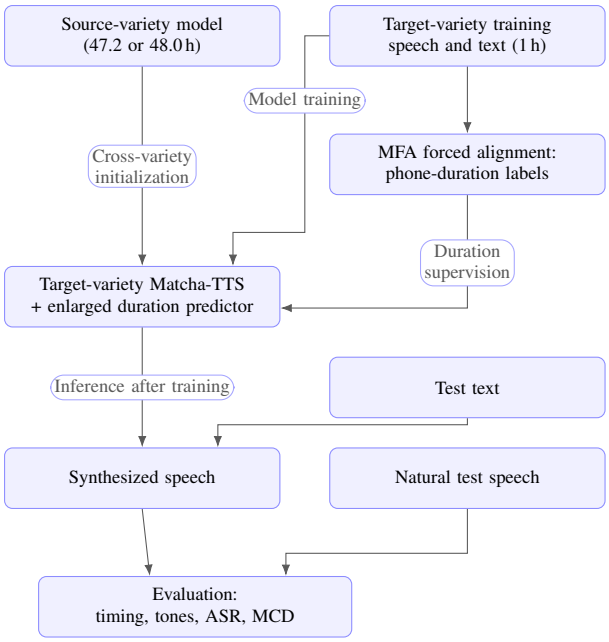


Fig. 1. Overview of XAD-TTS.

III. Experimental Setup

A. Data

The experiments use the TTS part of the Hakka Across Taiwan (HAT) corpus [10]: read speech in Sixian and Hailu,

each by one professional female and one professional male speaker. Utterances with code-switching or syllable contractions are excluded. Validation and test sets are defined by recording prompt, so their prompts never occur in training of the same variety. Training subsets of 1 h and 10 h are balanced across the two speakers and nested (the 10 h set contains the 1 h set). Pinyin is split deterministically into initial, vowel nucleus, and coda, and the tone is attached to the nucleus symbol ("joint" symbols). Section IV-A also uses "separate" symbols, in which tones are separate inputs, so 84% instead of 55% of the symbols are shared by the two varieties (symbols used by both varieties divided by symbols used by either).

B. Systems and training

Three systems are compared, all with joint symbols: Base, the original Matcha-TTS baseline trained from scratch with MAS durations; +AD, trained from scratch with alignment-supervised durations (AD) only; and XAD, the full XAD-TTS recipe, which adds cross-variety initialization and the enlarged predictor to +AD. Each is trained with Adam (learning rate 10^{-4}, batch 16) for a fixed 40,000 steps with three training seeds (1234, 123, 2026) at 1 h and 10 h, and the 40,000-step checkpoint is evaluated. As a high-resource reference, Base is also trained on the full pool (48.0 h of Sixian, 47.2 h of Hailu) with early stopping and a single seed. One 40,000-step run takes about 2 h on a single RTX 5070 Ti GPU (about 5.3 steps/s, 3.9 GB memory), and Base has 20.9 M parameters (XAD about 22.4 M because of the enlarged predictor). Synthesis uses 10 ODE steps, temperature 0.667, and the universal HiFi-GAN vocoder.

C. Evaluation

For each variety, 100 test sentences (50 per speaker) are randomly sampled from the test set and synthesized by every system. They are not filtered for text overlap with the other variety, whose training pool contains 15 of 411 distinct Sixian and 26 of 409 distinct Hailu test texts. 95% confidence intervals (CIs) are obtained by resampling sentences 5,000 times; with several seeds, every seed uses the same resampled sentences.

Spectral distortion

Mel-cepstral distortion (MCD) [11] is computed on 24 WORLD [12] mel-cepstral coefficients (without c_0) after dynamic time warping and averaged over five generation seeds per sentence.

Segment timing

Each synthesized waveform is force-aligned with the same MFA model, and the duration of every vowel nucleus is compared with the same nucleus in the natural recording. Timing accuracy is the share of nuclei within ± 30 ms (2.6 frames) of the natural duration, reported separately for checked and other syllables, together with the Pearson correlation ρ of checked-syllable nucleus durations. At 1 h, tolerances of ± 20 and ± 50 ms give the same ranking of Base, +AD, and XAD.

Tone recognition

A gradient-boosting classifier is trained on up to 3,000 natural utterances of the full training pool with acoustic features only: a five-point F0 contour of the nucleus in semitones relative to the speaker median, its mean and slope, log-duration, voicing ratio, and the F0 of neighboring syllables. Its accuracy on the synthesized syllables is reported, and its accuracy on all natural test recordings serves as a reference.

Speech recognition check

Omnilingual ASR [13] (1B LLM decoder, version 2, language code `hak_Latn`) transcribes Hakka in a romanization with tone diacritics and stop codas *-p*, *-t*, *-k*. Each natural recording and its synthetic versions are transcribed, and the transcripts are aligned word by word. The measures are the character error rate (CER) between the ASR transcript of synthesized speech and that of the corresponding natural recording, computed on Latin letters and tone marks with spaces, digits, and non-Latin characters removed and pooled over sentences, the retention of checked syllables (natural-transcript words ending in *-p*/*-t*/*-k* whose aligned synthetic word still does), the letter-similarity gap between checked and other words, and the tone-mark agreement of aligned words. Twenty of the 200 test sentences (12 Sixian, 8 Hailu) are excluded because the recognizer's output for the natural recording was mostly non-Latin text; the remaining 180 sentences are the same for all systems, and the results are pooled over the two varieties. Copy-synthesis, in which natural mel spectrograms are passed through the same vocoder, serves as a reference condition.

IV. Results

A. Cross-variety initialization

In a preliminary sweep over five data sizes from 1 to 48 h, initialization lowered the validation loss in 16 of 20 settings; the gain was largest at 1 h (0.08–0.13) and was close to zero under the full-data conditions (48.0 and 47.2 h), so the analysis focuses on 1 h. Table I compares MCD between training from scratch and cross-variety initialization with 1 h of target-variety data and one training seed. Each model is trained for 40,000 steps with validation every 1,000 steps, and the checkpoint with the lowest validation loss is evaluated; this differs from Section IV-B, where the 40,000-step checkpoint is evaluated. Δ is the MCD of the initialized model minus that of the model trained from scratch, computed per sentence from unrounded values, so it can differ by 0.01 from the difference of the rounded means. The 95% CI in brackets is obtained by resampling sentences 10,000 times within each speaker. Table I uses its own 100 test sentences without these overlapping texts, so its MCD values are not comparable with Table II.

TABLE I
EFFECT OF CROSS-VARIETY INITIALIZATION ON MCD.

Target	Symbols	Scratch	Init.	Δ [95% CI]
Sixian	joint	6.36	5.99	−0.36 [−0.39, −0.34]
Sixian	separate	6.17	6.08	−0.09 [−0.12, −0.06]
Hailu	joint	6.46	6.59	+0.13 [+0.09, +0.18]
Hailu	separate	6.48	6.22	−0.26 [−0.30, −0.22]

Initialization lowers MCD in three of four settings, by up to 0.36 dB, but raises it for Hailu with joint symbols (+0.13 dB), mostly for the male speaker (+0.35 dB vs. −0.08 dB). Initialization alone is therefore helpful but not uniformly so, and XAD-TTS pairs it with alignment supervision.

B. Main results at one hour

Table II summarizes the results. Checked and Other give the timing accuracy of checked and other syllables, ρ the duration correlation of checked syllables, ΔMCD the paired difference from Base (1 h), and Tone the tone-classification accuracy. The 1 h system results are averaged over three training seeds. The full-data Base systems, Base (48.0 h) and Base (47.2 h), use one training seed, while the natural-recording rows report the classifier accuracy on real speech.

TABLE II
RESULTS FOR THE 1 H SYSTEMS AND FULL-DATA REFERENCES.

System	Checked (%) ↑	Other (%) ↑	ρ ↑	ΔMCD (dB) ↓	Tone (%) ↑
Sixian					
Base (1 h)	71.0	63.7	0.535	—	64.3
+AD (1 h)	77.0	70.4	0.651	−0.12	69.0
XAD (1 h)	**81.3**	**74.0**	**0.738**	**−0.33**	**73.0**
Base (48.0 h)	81.1	73.9	0.767	—	—
Natural recordings	—	—	—	—	71.5
Hailu					
Base (1 h)	69.4	54.9	0.576	—	62.1
+AD (1 h)	77.7	58.9	0.676	−0.10	64.2
XAD (1 h)	**82.2**	**65.9**	**0.743**	**−0.24**	**67.3**
Base (47.2 h)	83.0	68.6	0.797	—	—
Natural recordings	—	—	—	—	67.1

Both +AD and XAD improve the measured duration accuracy over Base, and XAD provides further gains over +AD under the evaluated settings. Alignment supervision alone (+AD) raises the share of checked-syllable vowels within ±30 ms by 6.0 (Sixian) and 8.3 (Hailu) percentage points. XAD adds another 4.3 and 4.5 points, reaching 81.3% [77.6, 85.2] and 82.2% [78.0, 86.4] against 71.0% [66.9, 75.2] and 69.4% [65.3, 73.8] for Base, which is close to the high-resource references (81.1% and 83.0%) for this measure. Other syllables improve as well (63.7% to 74.0% and 54.9% to 65.9%), and the correlation of checked-syllable durations rises from 0.54 to 0.74 (Sixian) and from 0.58 to 0.74 (Hailu), still below the references (0.77 and 0.80). XAD lowers MCD by 0.33 [0.31, 0.35] and 0.24 [0.20, 0.28] dB, two to three times the reduction from +AD (0.12 and 0.10 dB), and the XAD reduction is observed for all three seeds. Tone-classification accuracy rises by 8.8 [7.7, 9.9] and 5.3 [4.4, 6.2] percentage points and XAD exceeds +AD by 4.1 and 3.2 percentage points. These paired differences are pooled over seeds and syllables and can differ by 0.1 from the rounded means in Table II. The classifier accuracy on XAD-generated speech is close to that on natural recordings (measured on all test sentences). This does not imply that the synthesized tones are more natural or more accurate than human speech.

C. Effect of data size

Fig. 2 compares checked-syllable duration accuracy at 1 h and 10 h of target-variety training data, with three training seeds per system. The full-data Base references use one training seed. For the 1 h systems and the full-data references, the 95% CIs span about ±4 percentage points.

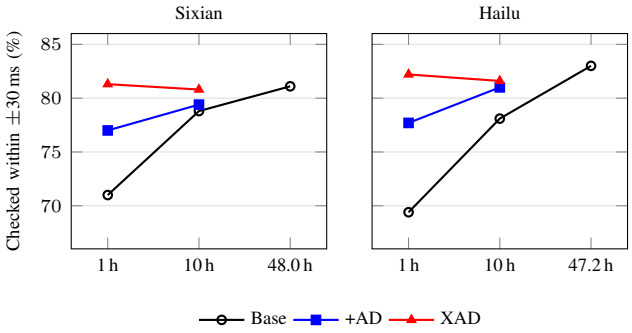


Fig. 2. Checked-syllable duration accuracy across data sizes.

The gains over Base are smaller at 10 h than at 1 h. At 10 h, XAD improves checked-syllable duration accuracy by 2.0 percentage points for Sixian and 3.5 percentage points for Hailu. It also reduces MCD by 0.24 dB and 0.16 dB, respectively, whereas +AD has MCD differences of +0.04 dB and +0.03 dB relative to Base. At 1 h, XAD achieves duration accuracy close to the full-data Base references on the ±30 ms measure; XAD additionally uses the source-variety model, so this comparison does not imply that only one hour of Hakka speech is involved.

D. Speech recognition check

Table III presents ASR-based results pooled across the two varieties using 180 test sentences. The measures include CER between the synthetic-speech and natural-speech ASR transcripts, checked-coda retention, the letter-similarity gap (Gap) between checked and other words, and tone-mark agreement. Copy-synthesis is included as a reference condition.

TABLE III
ASR-BASED EVALUATION RESULTS.

System	CER (%) ↓	Checked kept (%) ↑	Gap ↑	Tone marks (%) ↑
Base (1 h)	22.2	77.8	−4.1	73.9
+AD (1 h)	21.8	81.5	−1.5	76.7
XAD (1 h)	21.7	**82.6**	**−0.5**	77.1
Base (47.2/48.0 h)	19.6	80.1	−3.7	77.2
Copy-synthesis	9.8	89.4	+0.2	88.9

The ASR analysis provides an additional evaluation that does not directly use the MFA timing measurements or the tone classifier. Relative to Base, XAD changes CER by −0.5 percentage points with a 95% CI of [−2.7, +1.9], so the observed difference does not establish a clear CER improvement; the full-data Base reference has a lower CER (−2.6 [−4.5, −0.5]). The ASR transcripts retain checked codas more often (+4.8 percentage points [1.6, 8.0]), the checked-word similarity gap narrows from −4.1 to −0.5 (+3.6 [1.2, 6.0]), and tone-mark agreement increases by 3.2 percentage points [1.9, 4.4]. +AD alone already narrows the gap (+2.6 [0.4, 4.9]), whereas the full-data Base reference does not (+0.4 [−2.0, 2.9]) despite a similar tone-mark gain; this suggests that alignment-supervised durations, rather than the amount of target data alone, are associated with the checked-word improvement. XAD substantially reduces the checked-word similarity gap in this ASR-based analysis. A gap remains between XAD and the copy-synthesis reference in ASR transcript agreement (CER 21.7% vs. 9.8%); the factors contributing to this gap require further investigation.

E. Negative results

Several additions to XAD were compared directly with XAD: inference-time scaling of vowel durations (three seeds), a higher dropout rate in the duration predictor, pre-training of the duration predictor on source-variety durations, replay of source-variety data during target training (one training seed each), and averaging the predicted durations of the three seeds. None of them provided consistent improvements over XAD under the evaluated settings.

V. DISCUSSION AND LIMITATIONS

At 1 h of target-variety training data, +AD improves checked-syllable duration accuracy over Base (6.0 of the 10.3 percentage-point gain for Sixian and 8.3 of 12.8 for Hailu), and XAD provides further gains in the evaluated metrics. Because XAD changes both initialization and predictor capacity, the individual contributions of these two components remain to be isolated.

Several limitations should be considered. A listening test has not yet been completed, so the reported results are based on objective measures. The experiments use one TTS architecture and one vocoder. The method also requires a pronunciation lexicon and an in-domain aligner. The aligner was trained on all recordings of each variety, including the validation and test recordings, and therefore uses more target-variety speech than the one-hour TTS training subset; it saw only audio and transcripts, and no TTS model had access to validation or test audio. Because MFA is used for both duration supervision and timing evaluation, alignment errors may affect both stages, although all systems, including Base, are measured with the same aligner. In addition, the full-data Base reference uses one training seed, and in Table I the best validation checkpoint of every initialized model occurred at the first or second validation point (1,000 or 2,000 steps), so a finer validation interval might change these values. The ASR model and tone classifier are imperfect evaluation tools; on natural speech, the classifier reaches 71.5% accuracy for Sixian and 67.1% for Hailu.

VI. CONCLUSION

XAD-TTS combines cross-variety initialization, alignment-supervised durations, and an enlarged duration predictor. With 1 h of target-variety training speech, XAD-TTS improves checked-syllable duration accuracy, MCD, and objective tone-classification accuracy for both Sixian and Hailu. On the ±30 ms duration measure, its results are close to those of the full-data Base references. An additional ASR-based analysis shows improved agreement for checked-syllable codas and tone marks relative to the natural-speech transcripts. The aligner is not required during synthesis. A 40,000-step target-model training run takes approximately two hours on a single RTX 5070 Ti GPU; this time excludes source-model training, forced alignment, and data preparation. Future work includes listening tests and further experiments to examine the contribution of individual system components.

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